

Non-Invasive Anemia Detection

Estimating haemoglobin from photographs of the palpebral conjunctiva

Progress report · 20 August 2026

1. Summary

This period was spent building the image-processing and learning pipeline end to end: data loading and clinical labelling, conjunctiva extraction, quality control, the regression scaffold, an evaluation methodology, and a serving API. The guiding constraints are an eventually deployed screening app and small clinical cohorts — both punish silent failure, so the pipeline is engineered to produce numbers that can be defended: patient-grouped cross-validation, out-of-fold reporting only, and every result compared against a predict-the-mean baseline.

A real labelled cohort was obtained: 52 photographs of 26 children, both eyes, with laboratory haemoglobin and a full blood count. Screening it produced two findings. The pipeline processes real photographs without error and 94% pass quality control, but a plain colour-threshold extraction — which scores almost perfectly on synthetic images — proved **unreliable on real tissue**; a refined extractor was then built against these captures and now produces clean masks across all 52. Separately, the per-patient WHO threshold was validated against real laboratory values: a fixed single cutoff would have missed 15% of the anemic children in this cohort. Both are reported in Section 4.

Quantitatively, linear models over colour statistics were used to establish the floor a neural network must clear. On 26 patients **no arm beats a predict-the-mean baseline** (Section 6): mean colour alone carries no usable haemoglobin signal at this scale. The regression network has not been trained on real photographs, so **no accuracy figure in this report is a clinical result**.

1.1 Abbreviations

Defined here on first use; the full glossary is in GLOSSARY .md.

Term	Expansion	Term	Expansion
Hb	Haemoglobin (g/dL)	MAE	Mean Absolute Error
WHO	World Health Organization	MSE / RMSE	Mean Squared Error / its root
QC	Quality Control	R ²	Coefficient of determination
ROI	Region Of Interest	LoA	Limits of Agreement (Bland–Altman)
CIELAB	CIE L*a*b* colour space	TP / FP / TN / FN	True/False Positive/Negative
L* / a* / b*	Lightness / green–red / blue–yellow axes	PPV / NPV	Positive / Negative Predictive Value
CLAHE	Contrast Limited Adaptive Histogram Equalisation	F1	Harmonic mean of precision and recall
RGB / sRGB / P3	Colour encodings and profiles	Dice / IoU	Segmentation overlap measures
CNN	Convolutional Neural Network	CV / LOO	Cross-Validation / Leave-One-Out
ResNet	Residual Network (CNN architecture)	OLS	Ordinary Least Squares
CBC	Complete Blood Count	L1 / L2	Absolute / squared coefficient penalties

2. Pipeline design

The pipeline is a package of focused modules. The decision that matters most architecturally is that the inference path shares its preprocessing code with training but imports none of the training stack — so the

deployed server cannot drift from the conditions the model was trained under, and does not load training machinery to answer a request.

Area	Design
Extraction	Built around a redness prior in CIELAB refined by a seeded grabCut; all backends share one interface so a trained model swaps in without touching other stages
Quality control	Enforced, not merely recorded: failing captures are dropped from training; the API returns a retake prompt rather than a number
Clinical labelling	WHO thresholds applied per patient from age and sex
Evaluation	Patient-grouped, haemoglobin-stratified cross-validation; out-of-fold predictions only; always reported against a predict-the-mean baseline
Checkpoints	Self-describing bundle carrying weights, standardisation constants, preprocessing settings and metrics
Performance	Preprocessing cached to disk, so segmentation runs once rather than once per epoch
Data handling	Adapters for both public datasets behind a common record type; adding the hospital data requires one adapter
Deployment	FastAPI service loading one model at startup
Testing	19 automated tests, plus synthetic eye phantoms allowing development without data

4. Evaluation on real photographs

A local cohort of 26 children was obtained: both eyes photographed, giving 52 close-up captures of the everted lower eyelid, accompanied by laboratory haemoglobin, date of birth, gender, and a full blood count. The captures are well framed and in focus — tighter on the conjunctiva than typical field photography.

4.1 Cohort

Property	Value
Patients / images	26 / 52 (both eyes, exactly two captures each)
Haemoglobin	8.1 – 14.3 g/dL (mean 11.2, sd 1.37)
Anemic by WHO threshold	13 of 26 (50%)
Severe (Hb < 9.0)	3 patients
Age	3.4 – 16.7 years
Additional labels	Height, weight, socio-economic status, HCT, RBC, MCV, MCH, MCHC, RDW, platelets, MPV, TLC

Two properties make this cohort more useful than its size suggests. The 50/50 anemic split is unusually balanced — most cohorts skew heavily normal, which is what makes a predict-the-mean baseline hard to beat. And because both eyes of one child are photographed, the two captures are highly correlated: the loader assigns them a shared patient identifier so the grouped splitter cannot separate them; under a naive image-level split this dataset would produce substantially inflated results.

4.2 The pipeline runs on real data

Measure	Result
Images processed without error	52 of 52
Passed quality control	49 of 52 (94%)
Focus (Laplacian variance)	median 85.3, range 25.4 – 294.2
Mask coverage of frame	median 0.211, range 0.041 – 0.449

Measure	Result
Rejections	3, all blur (25.4, 34.2, 34.9 against a threshold of 35.0)

Two of the three rejections sit marginally below the threshold on captures that look usable to the eye. The blur threshold was chosen on synthetic images and appears slightly too aggressive for real photographs; it should be recalibrated once ground-truth masks make the downstream effect measurable.

4.3 Extraction: failure found, then fixed

Quality-control statistics say whether a capture is usable, not whether the segmentation found the right structure. Visual inspection of the overlays answers that, and the answer is unfavourable: on a substantial fraction of the 52 real captures the colour prior bleeds onto the eyelashes and lower-lid skin, and occasionally onto the sclera. On several it extends well below the lash line onto the cheek.

The cause is straightforward: eyelashes and peri-orbital skin are reddish-brown enough to pass a redness test, and neither is dark enough in absolute terms to be removed by a lightness gate.

A refined extractor was then built against these captures. It seeds a mask-initialised grabCut with what actually distinguishes the tissue: conjunctiva is smooth where the lash zone is high-frequency (a local-texture gate), sclera is bright and desaturated, iris and pupil are dark, and specular highlights sit on the wet tissue itself and are kept rather than discarded. Re-screened over all 52 captures, the refined extractor produces clean, tissue-hugging masks on visual review, and the median mask area halves — consistent with the bleed being removed rather than the tissue. The residual failure mode is a barely-everted lid, which is a capture problem rather than a segmentation one. The figure below shows the plain threshold (top) against the refined extractor (bottom) on the same captures.



Four real captures. Top row: the plain colour threshold, which bleeds onto lashes and skin. Bottom row: the refined seeded-grabCut extractor on the same images.

A method scoring 0.999 on synthetic images was visibly unreliable on real tissue — exactly the gap synthetic validation cannot reveal. The failure was characterised, a refined extractor was built against the real captures, and visual review across all 52 now shows clean masks. What remains missing is a *measured* Dice on real photographs, which requires the Eyes-defy-anemia ground-truth masks; until then the learned segmenter remains the primary long-term path.

4.4 The per-patient threshold validated on real data

A single fixed anemia cutoff (commonly 11.0 g/dL) is a tempting simplification, and this cohort measures exactly what it would cost, because its ages span three WHO threshold bands: 11.0 below five years, 11.5 from five to eleven, and 12.0 from twelve to fourteen. Twenty of the twenty-six children fall in the middle band.

Patient	Hb (g/dL)	Age	WHO threshold	Correct	Fixed 11.0 rule
10	11.0	11.1	11.5	anemic	normal
14	11.4	6.6	11.5	anemic	normal
16	11.3	8.4	11.5	anemic	normal
21	11.4	7.2	11.5	anemic	normal

A fixed 11.0 rule misclassifies 4 of 26 patients (15%), and every one is a false negative — an anemic child labelled normal. The errors cluster just below 11.5 because most of this cohort sits in the five-to-eleven band, where the correct cutoff is half a gram above the fixed value. This error occurs before the model makes a single prediction, so no amount of regression accuracy could recover it — which is why the pipeline predicts a continuous Hb and applies the threshold per patient.

5. Colour normalisation and lighting robustness

Lighting corrupts a photograph in two independent ways: a colour cast and a brightness shift. Two strategies address them. Gray-world white balance rescales the RGB channels until their means agree, correcting the *cast*. Applying contrast-limited histogram equalisation to the CIELAB lightness channel corrects *brightness* while leaving colour untouched; feeding the network only the a^*/b^* channels then makes it blind to brightness entirely.

5.1 Measured against laboratory haemoglobin

Both were scored on the local cohort using identical masks, so only the colour treatment differed. A structural point governs the comparison: equalising the lightness channel leaves a^*/b^* mathematically unchanged, so for the values the regressor consumes, that path is equivalent to no colour correction at all. The real contest is corrected versus uncorrected colour.

Measure	Uncorrected colour	Gray-world corrected
Correlation with laboratory Hb (Spearman)	+0.02	+0.30
Left-vs-right eye disagreement (median a^*)	2.69	2.02

Without colour correction the redness-haemoglobin relationship is essentially absent; per-shot illuminant differences swamp it. Correction recovers a modest but real association, and the same child's two eyes agree about 25% better.

5.2 Advantages and limitations of each approach

Property	Gray-world white balance	CIELAB (CLAHE on L^* , a^*/b^* input)
Corrects illuminant colour	Yes — the cast is equalised	No — the cast passes into a^*/b^* untouched
Corrects illuminant brightness	No	Yes — equalised, then discarded entirely
Preserves captured colour exactly	No — the shift depends on frame composition	Yes — a^*/b^* are mathematically unchanged
Invariance guarantee	None — a heuristic correction	Provable — brightness is absent from the input
Signal retained	Full RGB , including the pale-is-lighter cue	Colour only — the lightness cue is discarded
Robustness across devices	Better for colour; casts equalised	Better for exposure; weaker for colour

Property	Gray-world white balance	CIELAB (CLAHE on L*, a*/b* input)
Failure mode	Non-grey scenes skew the correction; a near-empty channel is amplified	Warm or cool lighting is read as physiology
Measured here	Spearman +0.30; eye gap 2.02; Hb SD 1.63	+0.02; 2.69; 2.74

The two correct **orthogonal** axes of the same problem, so they are complementary rather than competing. That motivated testing a hybrid — gray-world first, then a*/b* input — which is reported in Section 6.3. On present evidence gray-world alone is preferred, because discarding the lightness channel costs more information than the brightness invariance returns at this data scale.

5.3 Lighting stress test

One volunteer's eye was photographed under six illuminants (white, red, green, blue, magenta, purple). Because it is the same eye minutes apart, the true haemoglobin is constant and every reported difference is measurement error — no blood test is required for the test to be informative.

Spread across six illuminants (lower is better)	Gray-world	CIELAB
Redness (a*) standard deviation	17.58	32.76
Haemoglobin estimate standard deviation (g/dL)	1.63	2.74

Both figures are far too wide to be usable — a 4.5 g/dL swing from lighting alone spans the clinical range. Inspection of the masks explains why: **both pipelines segment correctly only under neutral light**, so the spread measures extraction collapsing as much as colour drift.

What this test design does well	What it cannot establish
The subject is constant, so the true haemoglobin is identical in all six frames and every reported difference is measurement error — no laboratory value is needed	One subject and one eye, so it characterises failure modes rather than accuracy
Isolates illumination as the only varying factor	The illuminants are far harsher than any clinical setting, so it maps where the method breaks, not whether it breaks in practice
Exposes a failure that neither synthetic images nor the cohort revealed	Segmentation collapse and colour drift are confounded, so the spread ratios are directional rather than precise

The underlying cause is physical, not algorithmic. In four of the six illuminants a colour channel is effectively dead — under blue light the red channel averages 0.9 of 255, under red and purple the green channel averages about 1. The lamp emitted nothing there, so the sensor recorded nothing there, and no normalisation can recover a measurement that was never made. Gray-world then divides by that near-zero mean and amplifies sensor noise. The practical remedy is a quality-control check on illuminant neutrality: the ratio of brightest to dimmest channel mean separates acceptable captures (1.7) from unusable ones (22–200) by two orders of magnitude.

6. Quantitative evaluation

How the haemoglobin figure is produced, and how well it performs. No library converts an eye photograph into a haemoglobin value; the libraries used supply a network architecture and weights pretrained on general photographs. The mapping from image to haemoglobin exists only because it is trained on paired examples, and it is the substance of the project.

Mechanically: the 224×224 crop is normalised to the statistics the pretrained weights expect, passed through a ResNet-18 whose classification head is replaced by a single-output linear layer, and the resulting scalar is converted to g/dL using the training mean and standard deviation stored inside the checkpoint. That value is then compared with the patient's WHO threshold.

6.1 Metrics reported

Metric	Why it is reported
MSE / RMSE / MAE	Squared error is the training objective; MAE is the figure a clinician reads as typical error
Median absolute error	Robust to a single catastrophic case, so divergence from MAE reveals outliers
Bias	Systematic offset; positive bias means over-estimating Hb, i.e. missing anaemia
R ²	Variance explained; zero means no better than predicting the mean, negative means worse
Bland–Altman limits of agreement	The standard clinical comparison of a device against a laboratory reference
Recall (sensitivity)	Fraction of anaemic patients caught — the error that sends an unwell patient home
Precision, specificity, NPV	Cost of false alarms, and trustworthiness of a negative result
F1, balanced accuracy	Summaries that resist the class imbalance which inflates plain accuracy

6.2 Synthetic phantoms — the machinery works

Metric	Model	Baseline
MAE (g/dL)	0.666	2.119
RMSE (g/dL)	0.836	2.499
R ²	0.888	0.000
Recall / Precision / F1	1.000 / 0.973 / 0.986	0.000 / — / 0.000

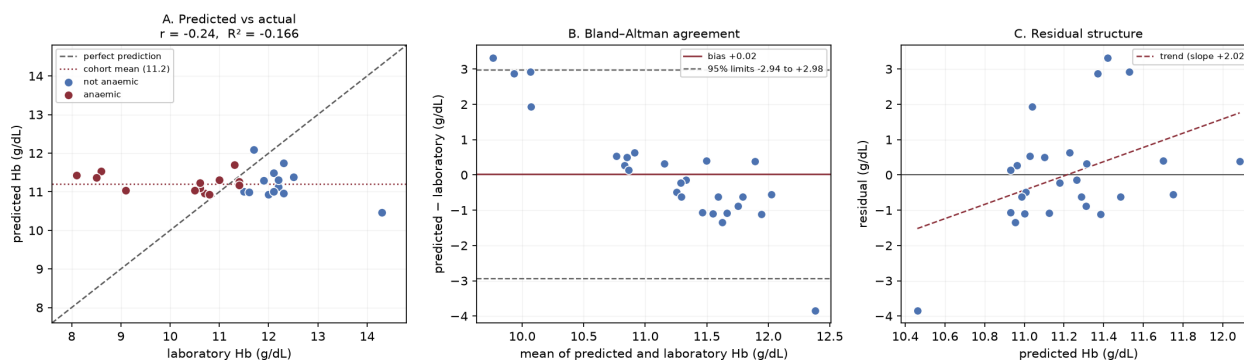
Reported for completeness and as verification of the metric code. The phantoms are flat-coloured shapes whose redness is a deterministic function of haemoglobin, so a strong score is expected by construction and carries no clinical meaning.

6.3 Real cohort — the linear floor is not yet met

No network has been trained on real data. The models below are deliberately **linear** — ordinary least squares and ridge regression over colour statistics — evaluated leave-one-patient-out. Their purpose is to establish the floor a neural network must clear: if a simple model already explains nothing, that bounds how much signal lies in summary colour and defines what the network has to find instead.

Arm	MAE	R ²	F1	Recall
End-to-end: gray-world + refined extractor, RGB	1.073	−0.166	0.688	0.846
End-to-end: CLAHE-L + own segmentation, a*/b*	1.135	−0.196	0.647	0.846
Shared mask: RGB representation	1.073	−0.166	0.688	0.846
Shared mask: a*/b* representation	1.192	−0.182	0.667	0.846
Shared mask: 24 features, plain least squares	11.444	−118.9	0.444	0.462
<i>Baseline: predict the training mean</i>	<i>1.083</i>	<i>−0.082</i>	<i>0.722</i>	<i>1.000</i>

The first pipeline wins on every metric in both framings, which is the fourth independent result in that direction, though at 26 patients the margin sits inside the noise. The decisive observation is different: **every arm has negative R² and an F1 at or below the trivial baseline**. Mean colour statistics from this cohort contain no usable haemoglobin signal.



Linear model on the real cohort, leave-one-patient-out. **A:** predictions cluster in a narrow horizontal band near the cohort mean while true values span 8.1–14.3 g/dL — the prediction spread is 23% of the actual spread, the visual signature of a model with no signal. **B:** Bland–Altman limits of -2.94 to $+2.98$ g/dL, far too wide to screen with. **C:** residuals rise steeply with the prediction (slope $+2.02$), the classic pattern of regression toward the mean.

Feature richness is not the constraint — data quantity is. Expanding from 3 features to 24 did not help the regularised model and destroyed the unregularised one: mean absolute error 11.4 g/dL and $R^2 -118.9$, a prediction wilder than the entire clinical range. With 24 parameters and 25 training points the model can nearly interpolate its training data, so it fits noise. This is the governing warning for the network stage, where roughly 8.4 million parameters will be trainable: the cohort must be pooled with the public datasets before any network result on real photographs can be believed.

6.4 Strengths and limitations of this evaluation

Strength	Limitation
Every metric is paired with an explicit baseline, so a model that has learned nothing cannot appear competitive	26 patients; confidence intervals comfortably exceed the differences between arms
Splits are grouped by patient, so two captures of one subject cannot straddle train and test	Leave-one-out gives a nearly unbiased but high-variance error estimate
Leave-one-out uses every patient as a test case, maximising scarce data	The regularisation strength is fixed rather than tuned, because tuning on the evaluation set would leak
Regression and screening are scored separately, so a good g/dL error cannot hide a poor clinical decision	Features are channel means; a network sees spatial structure that these summaries discard
The same code path scores synthetic and real data, so the comparison is like for like	One cohort, one device, one operator

7. Verification status

7.1 Verified

- All pipeline stages execute end to end; 19 of 19 automated tests pass.
- Cross-validated training completes on synthetic data: MAE 0.664 ± 0.096 g/dL, R^2 0.880, 1.45 g/dL better than a predict-the-mean baseline.
- A saved checkpoint carries everything needed to serve it, loads, and produces a haemoglobin estimate.
- Quality control correctly rejects blurred captures and returns a retake prompt.
- Clinical thresholds resolve correctly across age and sex.
- The pipeline processes real photographs end to end: 52 of 52 without error, 94% passing quality control.
- The per-patient WHO threshold is validated against real laboratory values: a fixed single cutoff misclassifies 15% of this cohort, all as false negatives (Section 4.4).
- A complete metric suite is implemented and reproducible — MSE, RMSE, MAE, median absolute error, bias, R^2 , Bland–Altman limits, accuracy, precision, recall, specificity, NPV, F1 and balanced accuracy —

each reported beside its baseline.

- Colour correction helps consistently: four independent measurements (cohort correlation, eye-to-eye consistency, lighting stress, representation ablation) all favour gray-world.

7.2 Measured and found wanting

- The plain colour threshold is unreliable on real tissue, bleeding onto eyelashes and lid skin. A refined seeded-grabCut extractor built against the same captures now produces clean masks on visual review across all 52 (Section 4.3); a measured Dice still awaits ground-truth masks.
- The blur threshold looks slightly too aggressive for real captures: two of three rejections were marginal on images that appear usable.
- Quality control does not yet test illuminant neutrality, so a strongly tinted capture passes and yields a confident but meaningless number.
- No linear model over colour statistics beats predict-the-mean on the 26-patient cohort (Section 6.3); the linear floor is unmet.

7.3 Not yet established

- Any *accuracy* on real photographs. The local cohort is labelled and could in principle supply one, but 26 patients is far too few for a defensible confidence interval, and segmentation on these images is known to be unreliable. Training on it is worthwhile only after the segmenter works.
- The segmentation model has not been trained — this requires the ground-truth masks.
- The CP-AnemiC loader has not been checked against a real download.
- No clinical validation of any kind has been performed.

8. Data

Two public datasets were identified and assessed, and a small labelled cohort is held locally. The professor has arranged hospital data, which will be the primary source; these sources allow development and benchmarking in the meantime.

Source	Images	Hb	Masks	Role
Eyes-defy-anemia	218	Yes	Yes	Only source of pixel-level ground truth; trains the segmenter
CP-AnemiC	710	Yes	No	Additional volume for the regressor
Local cohort	52	Yes	No	26 children, both eyes, with full blood count. Balanced 50/50 anemic. Too small to train on alone

Two consequences follow. First, roughly 900 images is a small corpus: it constrains model size, makes cross-validation necessary rather than optional, and means confidence intervals belong on every reported number. Second, the two sets differ in age, ethnicity, camera and lighting, so a model trained on one and evaluated on the other measures domain transfer as much as pallor. Results should be reported per source as well as pooled.

The local cohort carries a full blood count alongside haemoglobin. RDW and MCV distinguish iron-deficiency anemia from other types, which raises a possible extension: predicting which anemia rather than only whether. That is beyond current scope but worth recording. Provenance and licensing are unconfirmed, which matters if any figure derived from these images appears in a published write-up.

9. Risks

Risk	Mitigation
Small dataset produces a model that does not beat the mean baseline	Baseline comparison reported by construction, so this cannot go unnoticed

Risk	Mitigation
Colour heuristic proves inadequate on real photographs	Expected; the Mask2Former segmenter is the primary path and the heuristic only a fallback
Hospital data arrives in an unanticipated format	Loaders are isolated behind a common record type; an inspection command verifies parsing before training
Results overstated in the write-up	Grouped splits, out-of-fold reporting and baseline comparison are built into the pipeline rather than left to discipline

10. Next steps

#	Step	Outcome
1	Download both datasets and verify parsed counts against published figures	Confidence the loaders are correct
2	Train the segmentation model on Eyes-defy-anemia	First trained component, and the first measured Dice for the refined extractor
3	Benchmark all segmentation backends against ground truth	First real measurement in the project ; quantifies the rework
4	Train the regressor; report per-source and pooled with confidence intervals	First meaningful accuracy estimate
5	Add an illuminant-neutrality check to quality control, and recalibrate the blur threshold	Rejects tinted and marginal captures instead of scoring them
6	Pool the local cohort with the public datasets once segmentation is reliable	26 balanced, well-labelled patients added to training
7	Re-run Sections 6.2–6.3 on the trained network against the same baselines	First defensible accuracy claim on real photographs
8	Build the application around the existing API	Working demonstrator

Method record. Every approach tried, including those rejected, is recorded with its reason in `RESEARCH_LOG.md` — six method attempts that failed, ten defects found and fixed, four null results, and seven approaches deliberately not attempted.

Assessment. The pipeline is in a state where real data can be introduced and produce trustworthy numbers, has been exercised on real photographs rather than synthetic ones, and now reports a full metric suite against explicit baselines. Extraction — this period's focus — was taken from a failing colour threshold to clean masks across the full real cohort, with the remaining step being a measured Dice against ground-truth masks. The scientific question, whether conjunctival pallor predicts haemoglobin accurately enough in this cohort to be useful, remains open and is what the next phase must answer.